WARNING: Sorry, the genes arnD, arnE, fliR, gtrB, hcp, iagB, kdgM, kdgN, lptC, lptF, lptG, lpxA, mrdB, nanC, phoQ, prgI, rfaF, rfaQ, rfaY, rfbG, rfbN, rfbV, sigE, siiC, siiD, siiF, sopF, spaN, spiA, ssaD, ssaE, ssaI, ssaK, ssaL, ssaT, ssaU, ssas, sseE, sseK1, sseL, tagH, tisB, wbaV aren't in our dataset, which is focused on lead-operon promoters.
Please check out the other SalCom databases for our other comprehensive transcriptomic data.

Reference:

nature microbiology

Profiling *Salmonella* transcriptional dynamics during macrophage infection using a comprehensive reporter library

PDF (15 MB)

Welcome to SalComKinetics

SalcomKinetics is an interactive browser for the gene expression data presented in Temporal profiling of Salmonella transcriptional dynamics during macrophage infection using a comprehensive reporter library.
These data are collected from our GFP-reporter assay from 2,907 computationally identified promoter regions.

Instructions: Search by gene, locus or location (multiple entries separated by comma or space, use '*' for wild-card).

*To start: enter query or select Example: Top 100 genes in the early stage (1-4 h.p.i) • Top 100 genes in the middle stage (5-9 h.p.i) • Top 100 genes in the late stage (10-12 h.p.i) • Top 100 genes in the escape stage (13-15 h.p.i) • Entner-Doudoroff • Mn2+ regulation • SPI1 genes • SPI2 • SPI1 & 2 effectors • SPI3 • SPI5 • SPI6 • SPI1

arcB, arnD, arnE, bssR, csgD, fliR, gtrB, gtrB, hcp, iagB, invF, invH, kdgM, kdgN, nanC, ompL, lptC, lptF, lptG, lpxA, mgrB, m [Query]

Columns of the Heat Maps can be sorted by value - just click on the headers (45 ° rotated)

Heat map for expression of 32 genes in SL1344

Macrophage Expression Levels

	query	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	bssR	-	22.1	6.9	7.5	6.7	5.5	6	6.9	6.4	7.8	5.7	5.6	6.3	4.6	6.4
3	csgD	-	-	-	-	-	-	-	-	1.1	1.6	1.7	2.7	2.9	2.5	2.6
4	invF	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	invH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	ompL	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	mgrB	-	-	-	-	2.4	5	10	15.9	17.6	12.9	11.1	7.1	7.2	6.9	4.6
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	ompR	-	-	-	-	-	-	-	1.1	-	-	0.7	0.6	-	0.6	-
10	ompX	-	-	-	-	4	6.6	11	14.7	16	11.3	-	12.5	11.9	11	11
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	phoP	-	-	-	-	-	15.2	31.7	45.1	78.8	88.3	77	77	74.3	63.5	55.6
13	pipA	-	-	-	-	1.8	-	-	-	-	-	-	-	-	-	-
14	pipB	-	-	-	-	3.7	8.6	13.9	26.6	39.9	65.9	78.5	101.6	103.6	104.7	92.6
15	pipB	-	-	-	-	-	7.6	13.8	20.7	20.6	16.8	16.1	12.4	13.4	12	12.3
16	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	rfbU	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
21	rfc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	sipD	11.6	12.6	19.2	9.3	5.8	6.6	4.6	4.4	3.8	3.5	2.1	1.9	1.5	1.2	1.2
24	sopD2	7.7	5.9	-	-	2.7	4.2	5.3	6.7	10	10.7	11	7.6	7	7.6	7.7
25	sopE	-	-	-	-	-	1.8	1.6	2	-	-	-	-	-	-	-
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	spaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	ssaG	-	-	-	-	10.2	19.2	35.8	47.9	40.5	40.6	35.8	33.3	29.9	31.4	29.7
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	ssaR	-	-	-	4.1	15.1	24.1	37.4	45.4	36.4	38.3	34.4	28.7	33.1	26.8	26.5
31	sscB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	sseJ	-	-	-	-	10.1	22.7	33.4	51	54.8	49.7	38.2	44	42.8	42.8	34.2
33	steB	-	5.5	-	-	-	-	-	-	-	-	-	-	-	-	-
34	tagK	4.2	9.8	10.2	8	1.8	-	-	-	-	-	-	-	-	-	-
35	waaB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

	Gene_Plate	Gene	Synonym	Ty2 Accession	LT2 Accession	Paratyphi Accession	Category_manual	RSID	map2LT2	BioCyc SLIdentifier	BioCyc Common Name	BioCyc Gene Product
1	arcB_P14	arcB		T_RS16455	STM3328	SEEPA511_RS02580	Signal transduction	SL1344_RS17190	arcB	SL1344_3300		aerobic respiration two-component sensor histidine kinase ArcB
2	bssR_P27	bssR		bssR	STM0853	bssR	Biofilm	SL1344_RS04315	yliH	SL1344_0829	bssR	biofilm formation regulatory protein BssR
3	csgD_P11	csgD		T_RS09045	STM1142	SEEPA511_RS10375	fimbraie and pilus	SL1344_RS05625	csgD	SL1344_1079		transcriptional regulator CsgD
4	invF_P12	invF		T_RS14210	STM2899	SEEPA511_RS04905	virulence	SL1344_RS15005	invF	SL1344_2878	invF	AraC family transcriptional regulator InvF
5	invH_P12	invH		T_RS14215	STM2900	SEEPA511_RS04900	virulence	SL1344_RS15010	invH	SL1344_2879	invH	SPI-1 type III secretion system invasion lipoprotein InvH
6	ompL_P22	ompL		ompL	STM4016	ompL	Porin	SL1344_RS20630	yshA	SL1344_3962		porin
7	yobG_P8	yobG	mgrB	T_RS05300	STM1840	SEEPA511_RS13935	Signal transduction	SL1344_RS09190	yobG	SL1344_1769	mgrB	PhoP/PhoQ regulator MgrB
8	mrcA_P4	mrcA		T_RS20405	STM3493	SEEPA511_RS01720	Peptidoglycan	SL1344_RS18025	mrcA	SL1344_3460		carboxypeptidase/penicillin-binding protein 1A
9	ompR_P15	ompR		T_RS20360	STM3502	SEEPA511_RS01675	Signal transduction	SL1344_RS18070	ompR	SL1344_3469		DNA-binding response regulator
10	ompX_P21	ompX		ompX	STM0833	ompX	protein processing	SL1344_RS04215	ompX	SL1344_0808	ompX	outer membrane protease
11	pagP_P27	pagP						SL1344_RS03178				
12	phoP_P6	phoP		T_RS08600	STM1231	SEEPA511_RS10860	Signal transduction	SL1344_RS06080	phoP	SL1344_1169	phoP	two-component system response regulator PhoP
13	pipA_P27	pipA		T_RS09330	STM1087	SEEPA511_RS10060	virulence	SL1344_RS05340	pipA	SL1344_1026	pipA	type III secretion system effector protease PipA
14	pipB2_P12	pipB2		T_RS13590	STM2780	SEEPA511_RS05540	virulence	SL1344_RS14385	pipB	SL1344_2763	pipB2	SPI-2 type III secretion system effector PipB2
15	pipB_P7	pipB		T_RS09325	STM1088	SEEPA511_RS10065	virulence	SL1344_RS05345	pipB	SL1344_1027	pipB	SPI-2 type III secretion system effector PipB
16	rbsK_P16	rbsK		T_RS18950	STM3793	SEEPA511_RS00225	Carbohydrate Metabolism	SL1344_RS19525	STM3793	SL1344_3759	rbsK	ribokinase
17	EMPTY_P16	EMPTY		NaN	NaN	NaN	Carbohydrate Metabolism	SL1344_RS20000	rbsK	NaN	NaN	NaN
18	rbsK_P4	rbsK		T_RS18480	STM3885	SEEPA511_RS23035	Carbohydrate Metabolism	SL1344_RS20000	NaN	SL1344_3852		ribokinase
19	orf408_P31	orf408	rbsK	T_RS06865	STM1382	SEEPA511_RS06865	hypothetical	SL1344_RS06865	orf408	SL1344_1316		hypothetical protein
20	rfbU_P31	rfbU		T_RS03965	STM2086	SEEPA511_RS15250	LPS	SL1344_RS10730	wbaU	SL1344_2063	wbaU	GDP-mannose:α-L-Rha-(1→3)-α-D-Gal-PP-Und α-mannosyltransferase
21	rfc_P5	rfc		T_RS06195	STM1332	SEEPA511_RS11405	LPS	SL1344_RS06620	wzy_{O4}	SL1344_1266	wzyO4	O-antigen polymerase
22	siiA_P17	siiA		T_RS21165	STM4257	SEEPA511_RS21260	virulence	SL1344_RS21850	STM4257	SL1344_4193		hypothetical protein
23	sipD_P25	sipD		T_RS14130	STM2883	SEEPA511_RS04985	virulence	SL1344_RS14925	sipD	SL1344_2862	sipD	SPI-1 type III secretion system needle tip complex protein SipD
24	sopD2_P6	sopD2		T_RS09990	STM0972	SEEPA511_RS09390	virulence	SL1344_RS04725	STM0972	SL1344_0909	sopD2	SPI-2 type III secretion system effector SopD2
25	sopE_P13	sopE		T_RS21905		SEEPA511_RS20365	virulence	SL1344_RS13925		SL1344_2674	sopE	SPI-1 type III secretion system guanine nucleotide exchange factor SopE
26	sopE2_P8	sopE2		T_RS09265	STM1855	SEEPA511_RS14005	virulence	SL1344_RS09265	sopE2	SL1344_1784	sopE2	SPI-1 type III secretion system guanine nucleotide exchange factor SopE2
27	P30G1_P30	P30G1		spaM		SEEPA511_RS14180		SL1344_RS14180				
28	ssaG_P8	ssaG		T_RS06505	STM1406	SEEPA511_RS11735	virulence	SL1344_RS06985	ssaG	SL1344_1340		EscF/YscF/HrpA family type III secretion system needle major subunit
29	ssaM_P2	ssaM		T_RS06540	STM1413	SEEPA511_RS11770	virulence	SL1344_RS07020	ssaM	SL1344_1347	ssaM	SPI-2 type III secretion system apparatus protein SsaM
30	ssaR_P2	ssaR		T_RS06570	STM1419	SEEPA511_RS11800	virulence	SL1344_RS07050	ssaR	SL1344_1353	ssaR	SPI-2 type III secretion system export apparatus protein SsaR
31	sscB_P22	sscB		T_RS06490	STM1403	SEEPA511_RS11720	virulence	SL1344_RS06970	sscB	SL1344_1337		CesD/SycD/LcrH family type III secretion system chaperone
32	sseJ_P8	sseJ		T_RS08125	STM1631	SEEPA511_RS08125	virulence	SL1344_RS08125	sseJ	SL1344_1561	sseJ	SPI-2 type III secretion system effector SseJ
33	steB_P6	steB			STM1629		virulence	SL1344_RS26650	steB	SL1344_1559	steB	type III secretion system effector SteB
34	sciF_P21	sciF		T_RS13180	STM0271	SEEPA511_RS01370	virulence	SL1344_RS01370	STM0271	SL1344_0265	tagK	type VI secretion system-associated protein TagK
35	rfaB_P20	rfaB		T_RS19330	STM3719	SEEPA511_RS00605	LPS	SL1344_RS19160	waaB	SL1344_3685	waaB	"lipopolysaccharide α-(1,6)-galactosyltransferase"

≤ 0 10 ≥ 100

Promoter activity dynamics throughout intracellular macrophage infection. During the early stage of infection (1-4 h.p.i.), activity was quantified as the background-subtracted GFP signal. Promoter activity during the middle, late, and escape stages (5-15 h.p.i.) was quantified as the background-subtracted GFP signal, normalized by background-subtracted dTomato signal to account for intracellular replication. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For more details about normalization, please see Methods here.

download data:

***In Vitro* Expression Levels**

	query	00h InSPI2	01h InSPI2	02h InSPI2	03h InSPI2	04h InSPI2	05h InSPI2	06h InSPI2	07h InSPI2	08h InSPI2	09h InSPI2	10h InSPI2	11h InSPI2	12h InSPI2	13h InSPI2	14h InSPI2	15h InSPI2	16h InSPI2	17h InSPI2	18h InSPI2	19h InSPI2	20h InSPI2	21h InSPI2	22h InSPI2	23h InSPI2	24h InSPI2
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	bssR	2,787	2,323	4,103	6,290	9,833	16,923	22,114	27,513	29,933	31,552	28,428	22,943	18,428	14,629	12,000	11,800	11,600	11,400	10,800	10,800	10,900	10,900	10,200	10,500	9,780
3	csgD	30	73	180	673	1,213	2,467	4,753	6,530	7,676	7,129	8,029	7,661	6,751	5,153	4,110	4,060	4,140	3,690	3,770	3,940	3,550	3,660	3,550	3,620	3,620
4	invF	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	invH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	ompL	600	290	547	653	373	710	767	539	251	351	86	542	384	349	407	170	307	152	152	220	240	282	542	543	525
7	mgrB	207	300	287	640	937	1,393	1,887	2,026	1,888	1,866	1,733	1,975	1,679	1,123	1,200	781	510	859	829	749	704	791	848	805	889
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	ompR	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
10	ompX	327	200	513	413	1,420	3,850	7,790	13,980	15,086	16,067	16,227	17,139	15,920	11,445	9,270	7,780	7,700	7,290	7,590	7,680	7,720	7,810	7,960	8,000	8,350
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	phoP	23	-250	-130	253	1,760	3,033	5,823	7,559	7,519	6,940	7,505	7,198	5,936	4,326	3,560	3,240	2,960	3,100	3,200	3,290	3,260	2,890	3,140	2,900	2,880
13	pipA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	pipB	-310	-623	-670	-563	-667	-490	207	1,266	2,300	2,861	3,371	3,386	2,959	2,302	1,940	1,440	1,140	956	1,090	1,040	1,250	978	1,190	1,070	1,100
15	pipB	-247	-187	-103	-250	173	497	1,370	2,274	2,998	3,434	3,395	2,813	2,161	1,377	1,120	788	856	1,110	906	915	988	951	1,030	762	774
16	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	rbsK	450	413	290	650	737	1,023	935	655	656	740	544	404	234	339	438	394	470	398	346	366	207	244	497	240	332
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	rfbU	307	390	197	563	710	747	614	348	292	255	126	-173	76	-48	-94	-20	-22	29	-78	0	-124	-184	-105	-65	-183
21	rfc	370	293	110	297	247	33	577	499	805	765	693	744	901	826	590	264	594	657	230	281	367	529	370	330	150
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	sipD	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	sopD2	-630	-767	-690	-783	-620	-567	497	997	2,272	3,022	3,945	3,426	3,065	2,390	1,960	1,690	1,460	1,560	1,730	1,490	1,790	1,550	1,850	1,440	1,480
25	sopE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	spaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	ssaG	190	187	373	460	1,097	2,050	3,785	5,971	7,562	8,849	8,553	7,864	5,844	4,452	3,330	3,100	2,900	3,070	2,890	2,790	2,870	2,710	2,400	2,790	2,580
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	ssaR	-97	-230	43	107	560	1,577	4,507	5,774	6,967	7,815	7,464	6,477	5,264	4,132	3,220	3,220	3,090	3,050	3,300	3,250	3,150	3,470	3,200	3,000	3,170
31	sscB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	sseJ	-140	30	153	147	353	1,203	3,517	6,409	9,000	11,198	11,935	11,903	9,871	7,460	5,790	5,700	5,480	5,250	5,280	5,240	5,220	5,420	5,270	5,290	5,100
33	steB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
34	tagK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
35	waaB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

	query	00h InSPI2 low Mg	01h InSPI2 low Mg	02h InSPI2 low Mg	03h InSPI2 low Mg	04h InSPI2 low Mg	05h InSPI2 low Mg	06h InSPI2 low Mg	07h InSPI2 low Mg	08h InSPI2 low Mg	09h InSPI2 low Mg	10h InSPI2 low Mg	11h InSPI2 low Mg	12h InSPI2 low Mg	13h InSPI2 low Mg	14h InSPI2 low Mg	15h InSPI2 low Mg	16h InSPI2 low Mg	17h InSPI2 low Mg	18h InSPI2 low Mg	19h InSPI2 low Mg	20h InSPI2 low Mg	21h InSPI2 low Mg	22h InSPI2 low Mg	23h InSPI2 low Mg	24h InSPI2 low Mg
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	bssR	1,990	1,393	1,497	2,370	5,280	9,103	15,966	21,501	24,307	28,414	30,006	32,708	32,319	29,377	26,300	19,500	16,400	14,100	12,900	11,100	11,300	11,200	11,100	11,100	11,300
3	csgD	40	43	137	513	843	1,723	3,523	5,121	5,261	5,559	5,711	5,618	6,046	5,454	4,380	3,930	3,580	3,050	2,660	2,270	1,900	1,980	2,080	1,990	2,160
4	invF	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	invH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6	ompL	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	mgrB	100	150	313	467	1,547	2,470	3,593	3,751	3,421	2,984	2,867	2,856	2,806	2,636	2,090	1,780	1,590	1,220	1,100	1,060	1,190	881	1,030	1,160	948
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	ompR	100	183	260	453	790	1,150	1,303	1,259	965	843	959	886	593	466	271	225	40	143	149	106	207	84	211	206	2
10	ompX	-117	-227	-70	403	1,093	2,627	6,917	12,913	17,823	20,930	18,291	17,981	17,295	17,271	17,500	18,000	16,600	13,800	11,400	9,960	9,100	8,050	7,340	6,420	6,340
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	phoP	317	67	497	1,317	3,497	7,023	12,487	16,550	17,538	16,324	15,562	15,816	15,353	13,864	11,900	9,860	8,730	7,310	6,450	5,710	5,160	5,270	5,330	5,130	5,290
13	pipA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	pipB	110	-97	-67	10	70	273	1,153	2,273	3,018	3,740	4,265	4,393	4,896	5,065	4,900	4,560	4,210	3,590	3,090	2,660	2,240	2,050	1,690	1,600	1,420
15	pipB	-280	-207	-420	-317	317	1,400	3,144	4,600	5,178	4,721	4,756	4,312	3,307	2,524	1,980	1,480	1,340	1,060	955	1,140	1,070	1,160	1,010	1,170	1,060
16	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	rfbU	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
21	rfc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	sipD	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	sopD2	160	40	207	343	697	1,230	2,193	3,980	4,383	4,745	5,016	4,980	4,789	4,357	3,570	2,890	2,500	2,260	1,850	1,920	1,740	1,900	1,910	1,960	1,690
25	sopE	923	813	710	720	680	747	710	863	719	627	775	587	621	582	456	427	372	340	230	291	168	218	239	167	142
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	spaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28	ssaG	447	383	517	913	2,167	4,167	8,294	11,612	14,277	15,929	16,148	15,433	13,873	11,800	9,390	7,790	6,520	5,590	4,440	4,460	4,310	4,210	4,300	4,270	4,270
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	ssaR	-177	-243	-167	307	1,220	3,460	7,587	10,613	11,518	10,253	10,152	9,054	7,451	5,709	4,840	3,740	3,070	2,920	3,080	2,860	3,080	2,990	3,000	3,150	3,130
31	sscB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	sseJ	-3	-43	67	127	797	2,547	7,427	12,271	17,157	20,890	23,463	25,589	24,107	21,641	17,800	15,700	13,100	10,600	8,960	7,410	6,320	5,710	5,590	5,480	5,010
33	steB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
34	tagK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
35	waaB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

	query	00h NonSPI2	01h NonSPI2	02h NonSPI2	03h NonSPI2	04h NonSPI2	05h NonSPI2	06h NonSPI2	07h NonSPI2	08h NonSPI2	09h NonSPI2	10h NonSPI2	11h NonSPI2	12h NonSPI2	13h NonSPI2	14h NonSPI2	15h NonSPI2	16h NonSPI2	17h NonSPI2	18h NonSPI2	19h NonSPI2	20h NonSPI2	21h NonSPI2	22h NonSPI2	23h NonSPI2	24h NonSPI2
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	bssR	1,263	1,010	1,500	2,567	6,540	16,793	26,343	26,742	26,113	31,335	33,239	36,465	38,008	38,754	40,400	41,800	43,500	46,600	47,500	49,200	52,100	53,800	55,800	57,900	59,300
3	csgD	147	97	203	983	1,893	3,767	6,697	4,232	4,566	5,338	6,245	6,927	8,141	8,750	9,520	10,900	11,300	12,400	13,200	13,700	14,200	14,500	15,100	15,500	16,100
4	invF	-310	-360	-587	-517	-213	-160	-33	307	399	746	899	843	841	771	916	781	786	845	677	648	609	588	544	459	535
5	invH	-197	60	-277	-277	137	370	582	821	721	710	896	818	976	911	1,250	1,180	1,100	1,270	1,210	1,290	1,180	1,140	1,100	1,050	1,150
6	ompL	707	660	750	647	717	560	336	14	205	300	267	213	179	238	122	172	292	-18	150	156	243	140	107	66	106
7	mgrB	-173	123	-343	367	697	1,710	3,803	3,093	2,976	3,055	3,267	3,097	3,305	3,309	3,150	3,270	3,120	3,170	3,000	3,170	3,080	2,950	2,800	2,640	2,620
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	ompR	113	0	-333	-43	213	53	373	260	369	658	614	746	861	959	1,070	1,220	1,280	1,360	1,290	1,340	1,480	1,400	1,450	1,560	1,660
10	ompX	330	-150	247	763	1,597	4,653	10,043	17,517	15,290	15,291	16,357	16,550	17,683	18,092	18,600	18,900	19,800	20,100	20,200	20,800	21,000	21,200	21,700	21,500	21,700
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	phoP	650	447	363	1,170	3,393	7,140	22,513	21,912	19,101	18,960	19,517	19,991	19,951	19,427	19,600	19,100	19,300	18,800	18,200	17,300	17,000	16,900	16,700	16,200	16,300
13	pipA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	pipB	103	87	-193	57	67	77	-63	530	611	1,977	2,441	2,884	3,126	3,445	3,810	3,630	3,800	4,380	4,450	4,810	5,000	5,230	5,570	5,970	6,120
15	pipB	-213	-43	-100	-193	-173	-257	-340	-242	464	477	717	846	939	947	979	1,080	1,320	1,270	1,440	1,500	1,610	1,810	1,770	1,920	1,970
16	rbsK	-413	-207	-153	-223	-160	-360	-290	-241	408	1,329	1,959	2,332	2,856	3,505	3,670	4,070	4,550	4,570	4,970	5,270	5,650	5,890	6,220	6,340	6,520
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	rbsK	187	550	280	593	937	1,117	1,212	927	777	761	688	820	931	818	956	916	1,060	1,020	935	1,090	915	1,080	944	1,010	1,050
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	rfbU	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
21	rfc	170	90	177	127	-13	7	513	229	475	833	1,025	1,220	1,165	1,309	1,280	1,260	1,210	1,190	1,230	1,270	1,240	1,250	1,260	1,180	1,240
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23	sipD	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	sopD2	147	510	-33	153	477	490	330	333	475	595	971	818	822	975	1,190	1,210	1,320	1,450	1,350	1,400	1,540	1,580	1,480	1,600	1,900
25	sopE	-623	-333	-580	-503	13	840	1,261	1,615	1,380	1,761	1,802	2,190	2,245	2,100	2,080	2,160	2,210	2,170	2,100	2,130	1,870	1,820	1,870	1,820	1,670
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
27	spaM	87	90	13	93	407	290	533	511	285	812	1,231	1,507	1,982	2,228	2,540	2,760	2,880	2,930	2,890	3,050	3,170	3,070	3,310	3,390	3,720
28	ssaG	-67	0	-373	60	187	370	400	475	639	939	1,205	1,493	1,659	1,834	1,680	1,840	2,150	2,180	2,170	2,400	2,580	2,510	2,620	2,630	2,790
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30	ssaR	-173	-117	-3	-237	50	307	687	694	754	1,166	1,353	1,668	2,158	2,023	2,100	2,430	2,650	2,750	2,960	3,130	3,390	3,370	3,580	3,830	4,180
31	sscB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	sseJ	-150	17	-367	-230	-93	-137	-187	104	801	1,150	1,372	1,560	1,873	2,093	2,110	2,280	2,610	3,080	3,270	3,550	3,740	3,680	4,100	4,130	4,530
33	steB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
34	tagK	-310	-593	-440	-470	-357	-353	-410	-902	-574	416	312	367	318	614	673	710	728	826	921	913	1,060	1,000	1,040	1,060	957
35	waaB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

	query	00h 10% BHI	01h 10% BHI	02h 10% BHI	03h 10% BHI	04h 10% BHI	05h 10% BHI	06h 10% BHI	07h 10% BHI	08h 10% BHI	09h 10% BHI	10h 10% BHI	11h 10% BHI	12h 10% BHI	13h 10% BHI	14h 10% BHI	15h 10% BHI	16h 10% BHI	17h 10% BHI	18h 10% BHI	19h 10% BHI	20h 10% BHI	21h 10% BHI	22h 10% BHI	23h 10% BHI	24h 10% BHI	Gene_Plate	Gene
1	arcB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	arcB_P14	arcB
2	bssR	1,823	2,187	2,297	5,723	10,390	14,600	15,274	19,216	21,196	22,301	22,784	23,978	24,509	25,183	32,700	33,500	35,100	37,500	39,400	39,900	40,800	40,400	40,700	39,900	39,600	bssR_P27	bssR
3	csgD	1,143	1,423	1,293	2,137	3,839	4,988	6,183	8,123	8,593	8,800	9,254	8,992	9,199	8,589	8,390	8,720	8,750	8,410	8,220	8,020	7,390	7,340	7,100	6,940	6,860	csgD_P11	csgD
4	invF	193	470	770	1,443	3,792	4,260	4,381	3,815	3,262	2,723	2,385	2,014	1,931	1,834	1,410	1,650	1,580	1,560	1,600	1,580	1,470	1,400	1,410	1,320	1,250	invF_P12	invF
5	invH	400	463	530	1,020	1,648	1,774	1,799	1,980	1,944	1,689	1,693	1,529	1,608	1,385	1,340	1,350	1,340	1,400	1,390	1,330	1,060	1,110	1,020	1,120	834	invH_P12	invH
6	ompL	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ompL_P22	ompL
7	mgrB	567	523	1,047	1,693	2,358	2,532	2,790	2,907	2,616	2,436	2,462	2,297	2,169	1,919	1,820	2,110	2,170	2,010	1,920	1,800	1,920	1,740	1,760	1,660	1,610	yobG_P8	yobG
8	mrcA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	mrcA_P4	mrcA
9	ompR	1,340	1,180	830	780	1,276	741	1,221	1,449	1,474	1,675	1,624	1,730	1,609	1,547	1,410	1,480	1,490	1,310	1,040	1,150	842	881	787	646	640	ompR_P15	ompR
10	ompX	297	1,003	2,267	5,003	8,757	12,421	13,651	16,134	18,567	21,339	23,541	25,548	26,957	27,726	27,900	28,100	29,800	30,300	29,400	28,300	27,800	27,100	26,800	26,000	25,200	ompX_P21	ompX
11	pagP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	pagP_P27	pagP
12	phoP	407	963	3,240	9,100	13,704	15,056	16,081	14,513	13,943	13,672	13,355	12,723	11,853	10,992	10,500	11,500	11,900	11,800	11,200	11,000	10,700	10,300	10,300	9,760	9,670	phoP_P6	phoP
13	pipA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	pipA_P27	pipA
14	pipB	320	383	183	407	1,213	1,610	2,134	2,792	3,796	4,552	5,160	5,814	6,489	7,089	7,140	7,940	8,810	9,380	9,810	9,750	9,960	9,930	9,790	9,620	9,530	pipB2_P12	pipB2
15	pipB	-437	-593	-310	13	80	234	407	694	1,136	1,542	2,032	2,853	2,919	3,097	3,690	3,560	3,440	3,080	2,860	2,880	2,590	2,490	2,200	2,120	2,080	pipB_P7	pipB
16	rbsK	153	513	403	2,246	3,267	3,366	3,989	3,744	3,650	3,536	3,592	3,325	3,132	3,062	4,320	4,130	3,750	3,570	3,370	3,110	3,010	2,990	2,840	2,850	2,660	rbsK_P16	rbsK
17	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EMPTY_P16	EMPTY
18	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rbsK_P4	rbsK
19	rbsK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	orf408_P31	orf408
20	rfbU	1,193	1,247	1,023	1,357	1,355	1,069	1,185	846	715	852	813	670	656	675	614	632	417	525	289	453	242	326	246	236	237	rfbU_P31	rfbU
21	rfc	270	370	603	823	1,167	1,038	791	624	632	730	764	774	691	642	625	818	828	746	811	629	613	708	681	562	499	rfc_P5	rfc
22	siiA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	siiA_P17	siiA
23	sipD	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sipD_P25	sipD
24	sopD2	113	340	123	387	697	627	947	986	1,250	1,400	1,574	1,874	2,152	2,237	2,460	2,500	2,730	2,790	2,930	2,830	2,690	2,620	2,590	2,570	2,470	sopD2_P6	sopD2
25	sopE	750	1,090	1,220	2,703	6,730	9,232	9,367	9,705	8,876	8,703	8,012	7,604	6,997	6,254	6,270	7,410	6,690	6,320	5,930	5,670	5,690	5,460	5,220	4,970	4,770	sopE_P13	sopE
26	sopE2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sopE2_P8	sopE2
27	spaM	-1,127	-970	-937	-370	-204	249	324	498	494	634	562	550	740	691	1,080	1,010	1,230	977	869	693	770	729	735	585	755	P30G1_P30	P30G1
28	ssaG	40	0	473	560	1,060	425	1,066	1,711	2,639	3,872	5,285	6,520	7,452	8,350	10,400	12,200	12,500	13,000	13,500	13,600	13,700	13,000	12,600	12,100	11,900	ssaG_P8	ssaG
29	ssaM	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ssaM_P2	ssaM
30	ssaR	620	560	290	943	1,620	1,369	1,565	1,738	2,713	3,750	4,803	6,059	7,251	8,603	10,100	10,400	11,600	11,800	11,700	11,700	11,500	11,700	11,000	10,700	10,400	ssaR_P2	ssaR
31	sscB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sscB_P22	sscB
32	sseJ	283	193	397	287	409	321	796	2,444	4,462	6,995	9,544	12,109	14,301	16,748	25,500	30,200	32,400	33,400	33,300	34,100	33,900	33,800	33,700	33,300	33,500	sseJ_P8	sseJ
33	steB	-180	-83	-120	-313	347	297	560	767	978	906	847	629	794	698	721	628	623	604	712	642	591	489	528	516	521	steB_P6	steB
34	tagK	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	sciF_P21	sciF
35	waaB	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	rfaB_P20	rfaB

≤ 0 10k ≥ 100k

Promoter activity dynamics throughout growth in in vitro conditions (InSPI2, InSPI2 low Mg, NonSPI2, and 10% BHI). Activity was quantified as the background-subtracted GFP signal, normalized by background-subtracted OD600 to account for changes in cell density. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For visualization, one datapoint is shown per hour. For all of the dynamic data, please toggle to full resolution view to visualize all data points, or download the raw data. For more details about normalization, please see Methods here.

download data:

Results provided by the Huang and Monack Labs, automatically generated on Wed Nov 19 05:19:08 2025